WORKED SOLUTION

STEP II 2019, Question 4

Pure • Telescoping Trigonometric Products

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Read the question before you go on. It is in the STEP II paper in OCR's download of the 2019 STEP papers (a zip file):

www.ocr.org.uk/Images/696590-2019-past-papers.zip

The key idea

A product of cosines whose angles double (or halve) each time telescopes if you multiply by the sine of the smallest angle: the double-angle formula $2\sin\theta\cos\theta = \sin 2\theta$ then swallows the factors one at a time. Once the product is in closed form, taking logarithms turns it into a sum, and differentiating turns $\ln\cos$ into $\tan$: that is where the tangent sum comes from. The infinite versions then follow by letting $n \to \infty$ with the two standard limits.

Solution

Part (i)

Call the product P. Since $\sin\frac{\pi}{9} \neq 0$, we may multiply by it and apply the identity three times:

$$\begin{aligned} 8\sin\tfrac{\pi}{9}\,P &= 4(2\sin\tfrac{\pi}{9}\cos\tfrac{\pi}{9})\cos\tfrac{2\pi}{9}\cos\tfrac{4\pi}{9} = 2(2\sin\tfrac{2\pi}{9}\cos\tfrac{2\pi}{9})\cos\tfrac{4\pi}{9} \\ &= 2\sin\tfrac{4\pi}{9}\cos\tfrac{4\pi}{9} = \sin\tfrac{8\pi}{9} = \sin(\pi - \tfrac{\pi}{9}) = \sin\tfrac{\pi}{9}. \end{aligned}$$

Dividing by $8\sin\frac{\pi}{9}$,

$$\cos\tfrac{\pi}{9}\cos\tfrac{2\pi}{9}\cos\tfrac{4\pi}{9} = \frac{1}{8}.$$

Part (ii)

Here the angles $x, \frac{x}{2}, \ldots, \frac{x}{2^n}$ halve, so the smallest is $x/2^n$. As $0 < x/2^n < \frac{1}{2}\pi$, its sine is non-zero. Multiplying by $\sin(x/2^n)$, the factor $\cos(x/2^n)$ pairs with it to give $\frac{1}{2}\sin(x/2^{n-1})$, which pairs with $\cos(x/2^{n-1})$, and so on up to $\cos x$. Each of the $n+1$ steps contributes a factor $\frac{1}{2}$:

$$\sin\left(\frac{x}{2^n}\right)\prod_{k=0}^{n}\cos\left(\frac{x}{2^k}\right) = \frac{1}{2^{n+1}}\sin 2x, \qquad \text{so} \qquad \prod_{k=0}^{n}\cos\left(\frac{x}{2^k}\right) = \frac{\sin 2x}{2^{n+1}\sin(x/2^n)}.$$

Ask yourself How can differentiation turn a product of cosines into a sum of tangents?

Take logarithms. Every factor is positive, because $0 < x/2^k < \frac{1}{2}\pi$ for each k, and $\sin 2x > 0$ too. So

$$\sum_{k=0}^{n} \ln\cos\left(\frac{x}{2^k}\right) = \ln\sin 2x - (n+1)\ln 2 - \ln\sin\left(\frac{x}{2^n}\right).$$

Differentiate with respect to x, using $\frac{\mathrm{d}}{\mathrm{d}x}\ln\cos(x/2^k) = -\frac{1}{2^k}\tan(x/2^k)$ by the chain rule:

$$-\sum_{k=0}^{n} \frac{1}{2^k}\tan\left(\frac{x}{2^k}\right) = 2\cot 2x - \frac{1}{2^n}\cot\left(\frac{x}{2^n}\right).$$

Multiplying by -1 gives exactly the required result:

$$\sum_{k=0}^{n} \frac{1}{2^k}\tan\left(\frac{x}{2^k}\right) = \frac{1}{2^n}\cot\left(\frac{x}{2^n}\right) - 2\cot 2x.$$

Part (iii)

Dividing the product formula by $\cos x \neq 0$ and using $\sin 2x = 2\sin x\cos x$,

$$\prod_{k=1}^{n} \cos\left(\frac{x}{2^k}\right) = \frac{2\sin x\cos x}{2^{n+1}\sin(x/2^n)\cos x} = \frac{\sin x}{2^n\sin(x/2^n)}.$$

Ask yourself What happens to $2^n\sin(x/2^n)$ as $n \to \infty$?

Put $\theta = x/2^n$, so that $\theta \to 0$ as $n \to \infty$. Then

$$2^n \sin\left(\frac{x}{2^n}\right) = x \cdot \frac{\sin\theta}{\theta} \longrightarrow x \cdot 1 = x,$$

and therefore

$$\prod_{k=1}^{\infty} \cos\left(\frac{x}{2^k}\right) = \lim_{n\to\infty} \frac{\sin x}{2^n\sin(x/2^n)} = \frac{\sin x}{x}.$$

For the series, substitute $j = k+2$: then $\frac{1}{2^{j-2}} = \frac{1}{2^k}$ and $\frac{\pi}{2^j} = \frac{\pi/4}{2^k}$, so the sum is the series of part (ii) with $x = \frac{\pi}{4}$, which lies in $(0, \frac{1}{2}\pi)$. With this x, $2\cot 2x = 2\cot\frac{\pi}{2} = 0$, so the partial sum up to $k = n$ is

$$\frac{1}{2^n}\cot\left(\frac{\pi}{4\cdot 2^n}\right) = \frac{1}{x}\cdot\frac{\theta}{\tan\theta} \qquad \left(\theta = \frac{x}{2^n},\ x = \frac{\pi}{4}\right).$$

As $n \to \infty$, $\theta \to 0$ and $\theta/\tan\theta \to 1$, so the partial sums tend to $1/x$:

$$\sum_{j=2}^{\infty} \frac{1}{2^{j-2}}\tan\left(\frac{\pi}{2^j}\right) = \frac{4}{\pi}.$$

Checks

- Part (i): $\cos 20^\circ \cos 40^\circ \cos 80^\circ \approx 0.9397 \times 0.7660 \times 0.1736 \approx 0.1250$.
- Part (ii) with $n = 0$: the product is $\cos x$ and the formula gives $\sin 2x/(2\sin x) = \cos x$. The sum identity with $n = 0$ reads $\tan x = \cot x - 2\cot 2x$, which is the double-angle formula for cot rearranged.
- Part (iii): the first terms of the series are $\tan \frac{\pi}{4} = 1$, $\frac{1}{2}\tan \frac{\pi}{8} \approx 0.2071$, $\frac{1}{4}\tan \frac{\pi}{16} \approx 0.0497$, …; the total, confirmed by computer, is $1.27324\ldots = 4/\pi$.
- Letting $x \to 0$ in $\prod \cos(x/2^k) = \sin x/x$ gives $1 = 1$, as it should.

Common slips

- Counting the factors of $\frac{1}{2}$ wrongly: the product from $k = 0$ to n has $n + 1$ factors, so the denominator is 2^{n+1}, not 2^n.
- Taking logarithms without noting that every factor is positive; this is why the range $0 < x < \frac{1}{2}\pi$ matters.
- Forgetting that $2\cot 2x$ vanishes at $x = \frac{\pi}{4}$, or using $x = \frac{\pi}{2}$ (the $j = 1$ term would be $\tan \frac{\pi}{2}$, which is undefined).

Beyond STEP

Setting $x = \frac{\pi}{2}$ in $\prod \cos(x/2^k) = \sin x/x$ and writing each cosine with nested square roots gives Viète's formula $\frac{2}{\pi} = \frac{\sqrt{2}}{2} \cdot \frac{\sqrt{2+\sqrt{2}}}{2} \cdots$, the first infinite product in the history of mathematics. Differentiating the logarithm of an infinite product term by term, as we did with the finite one, needs justification; a first course in analysis sets out when it is allowed. Euler's product $\sin x = x\prod_{n\geq 1}\left(1 - \frac{x^2}{n^2\pi^2}\right)$ grows out of the same ideas. For the rigorous background, try Abbott, *Understanding Analysis*.

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